ColdTrust

An AI-assisted cold-chain guardian that checks its own sensor data before it acts.

ColdTrust is a working prototype that monitors a refrigerated truck's cargo hold, checks that its own sensor data can be trusted, and only then decides, explains and acts. Rules set each package's status, three AI agents explain the risk and propose actions, and an administrator approves anything disruptive.

Supporting evidence pack for the Technical Proposal by Team Bibilabu

Page	What it shows
2	Problem, users and solution, as in the Technical Proposal, with where each is evidenced
3	System architecture
4	The three AI agents and where the rules and the human stay in charge
5	The hardware prototype
6	Measured result: ColdTrust against a plain threshold alarm
7	Evaluation of the AI agents (60 live calls)
8	Dashboard: a faked-data attack is caught (screenshots)
9	Dashboard: suspected theft, automatic report and evidence (screenshots)
10	Sample incident report produced by the system (Word file from the team's template)
11	GS1 EPCIS 2.0 event record, with hash check
12	Automated tests, safeguards and limits

Read this first. Every scenario, number and picture in this pack comes from the built-in simulator or is a labelled sample frame, not from a real delivery. The simulator, the rules and the baseline were written by the same person, so the results show what the system does on known cases, not how often real incidents happen. Run it yourself: `judge_demo.bat` (README, no hardware, no sign-in).

Problem, users and solution

Problem

Temperature-sensitive products, including pharmaceuticals, vaccines and perishable goods, need reliable monitoring and timely responses to changes throughout the cold chain.

Research on cold-chain logistics in a developing Asian country found **temperature fluctuations**, **reliance on data loggers and manual monitoring**, and **insufficient expertise** as challenges (Abu Hassan, Ngah & Ab Talib, 2021). These methods can cause **unnecessary alarms** and **delayed responses to cooling-system failures**, and cannot explain the cause of an alert.

The Cybersecurity and Infrastructure Security Agency (CISA, 2020) highlights **vehicle hijacking and cargo tampering** as security risks and the need to **protect monitoring data against manipulation**.

Target users

Administrator: monitors shipments, reviews risks, approves actions, manages incidents and generates reports.

Driver: receives operational alerts and sees shipment information through a read-only interface.

Solution in one line

Sensor-data verification, product-specific risk assessment, AI-assisted incident analysis and human-approved responses. **Rules decide the status, the AI only explains and proposes, and a person approves anything disruptive.** Every number in AI text is checked against the data before it is shown.

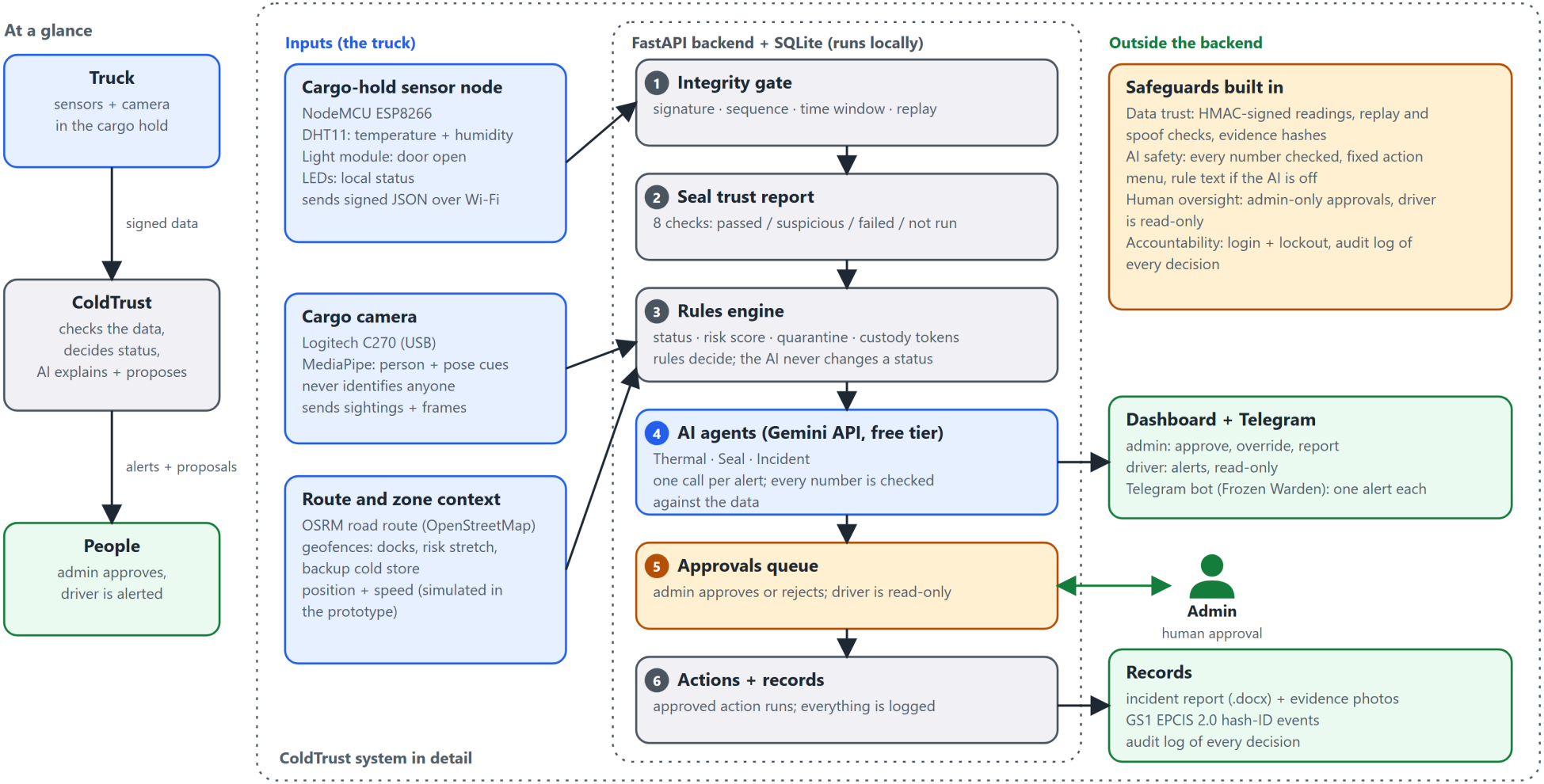
Challenge in the proposal	How ColdTrust responds	Evidence in this pack
Temperature fluctuations, unnecessary alarms, delayed response to cooling failures	Each product is judged against its own limits and excursion budget; the Thermal agent reviews the trend, and a maintenance report follows only when enough relevant readings exist	p6: 0 false alarms against 1; first alert 11 min earlier on a slow cooler failure
Alerts that cannot be explained; insufficient expertise	Three Gemini agents explain each alert in plain language and propose actions from a fixed menu; rule-based text is the fallback	p7 (60 live calls, 0 invented numbers); p8
Vehicle hijacking and cargo tampering	Camera finds a person in the hold without identifying anyone; the rules quarantine the cargo for admin review; a draft incident report with hashed evidence is prepared automatically	p9, p10
Monitoring data manipulated	HMAC signature, sequence, freshness and replay checks, then the Seal trust report (passed / suspicious / failed / not run)	p6: 3 of 3 data attacks recognised; p8
Traceability across the supply chain	Events are written as GS1 EPCIS 2.0 records with cryptographic hashes	p11

Problem text follows the submitted Technical Proposal. Reference: Abu Hassan, M. F., Ngah, A. H., & Ab Talib, M. S. (2021). The state of cold logistics supply chain in a developing Asian country: a preliminary insights. Operations and Supply Chain Management, 14(4), 467-475. CISA (2020), cold-chain security guidance.

System architecture

Architecture

ColdTrust: a cold-chain guardian that checks its own data before it acts

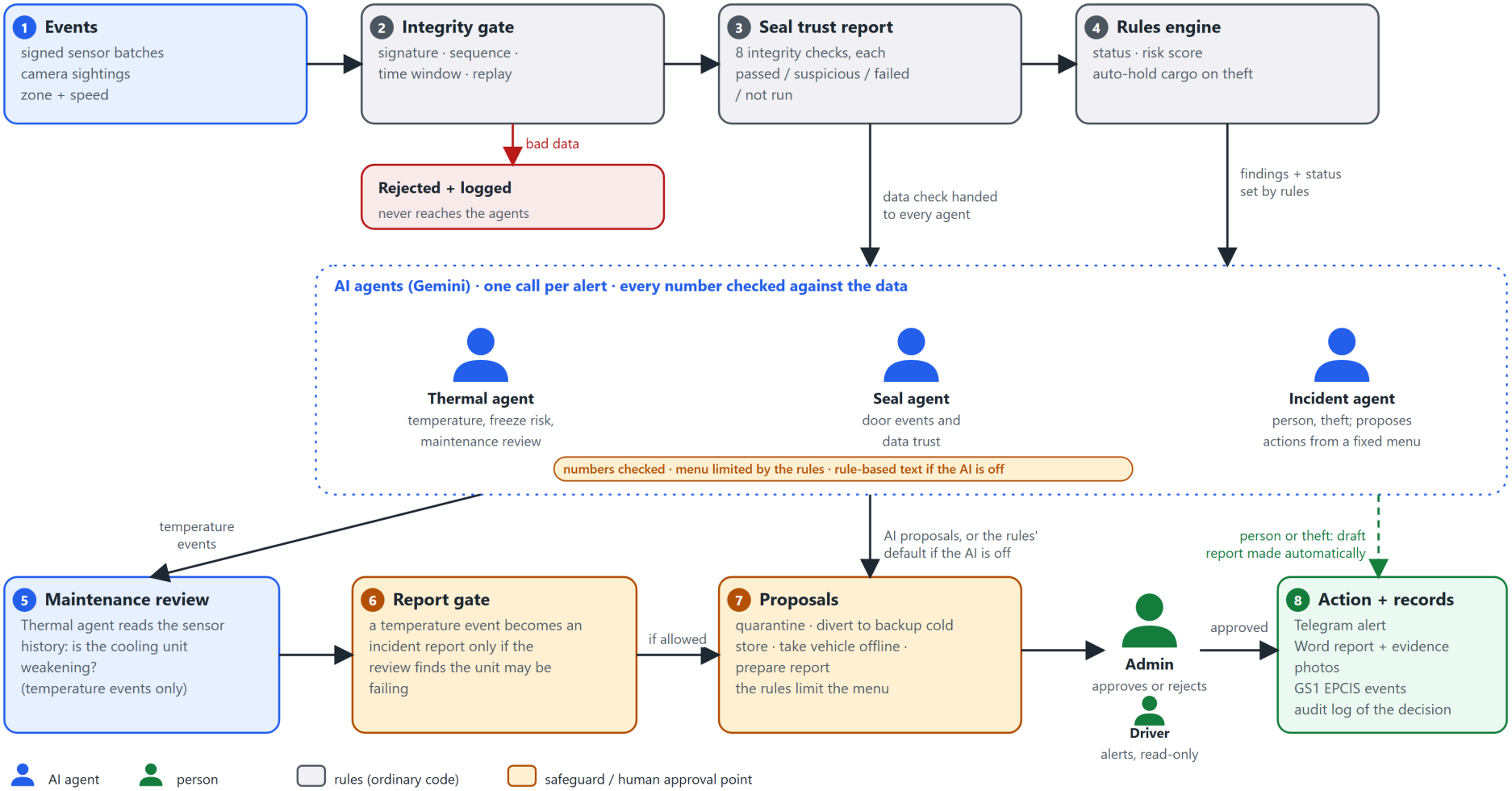


Source: docs/diagrams/architecture.svg (generated by tools/make_diagrams.py).

The three agents, the rules and the human

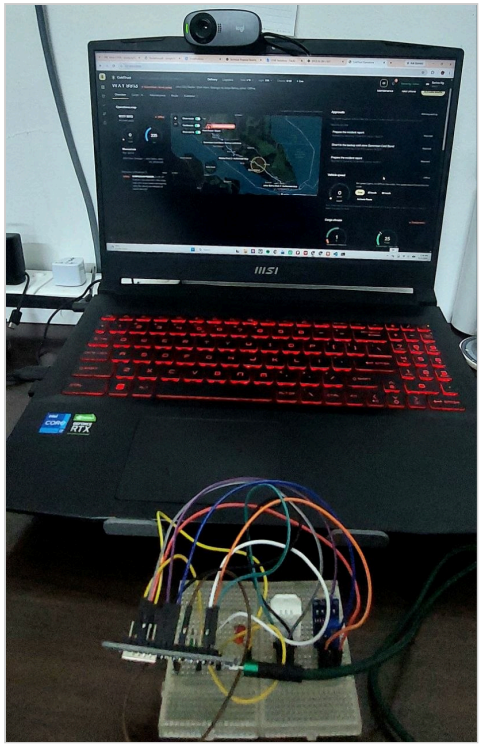
AI agent workflow

Rules decide the status. Agents explain and propose. A human approves anything disruptive.

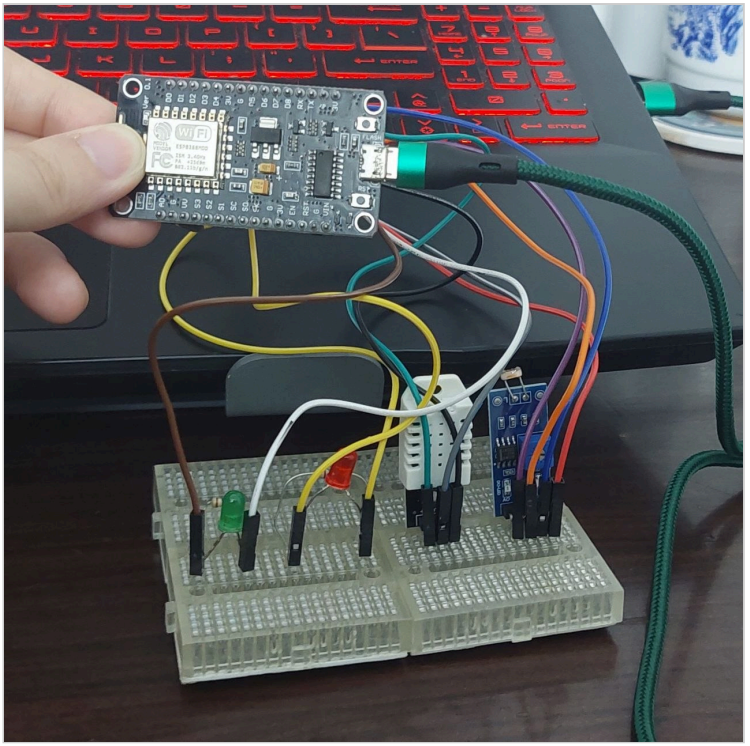


Source: docs/diagrams/agent_workflow.svg. **Thermal agent:** explains temperature deviations and possible cooling-system failure; a maintenance report is generated only when enough relevant readings exist. **Seal agent:** analyses data-integrity issues (invalid signatures, replays, contradictory readings). **Incident agent:** evaluates thermal and security risk, recommends actions from a fixed menu, and drafts an incident report automatically for theft or a suspicious person. The agents cannot change a package status or run a disruptive action; the administrator approves, edits or rejects.

Hardware prototype



Overall setup: hardware and software



Hardware only

Part	Role
ESP8266 NodeMCU	Reads the sensors every 5 s and sends readings in batches with an HMAC for backend verification (Wi-Fi)
DHT11	Cargo temperature and humidity
Light module	Light intensity inside the hold; a change approximates a door-opening event
Green and red LEDs	Green: heartbeat, reading sent. Red: alarm or Wi-Fi down
Logitech C270 webcam	MediaPipe person detection in the cargo compartment, without identifying individuals; photos stored with cryptographic hashes

Firmware and wiring: firmware/README.md. The device key is in firmware/secrets.h (git-ignored). The DHT11 reads whole degrees only; the second thermometer is not fitted and GPS is simulated, which the dashboard labels.

Measured result: ColdTrust against a plain threshold alarm

Measure	Plain threshold alarm	ColdTrust
False alarms (normal driving and a dock stop)	1	0
Data attacks recognised (replay, flat spoof, contradicting sensor)	0 of 3	3 of 3
Incidents where cargo was already held at the first alert	0 of 5	5 of 5
Slow cooler failure: first alert	19:00	8:00 (11 min earlier)

Per scenario (minutes:seconds from the start of the simulated run)

Scenario	What happens	Alarm	ColdTrust alert	ColdTrust says	Held at first alert
normal	Six quiet minutes	-	-	-	0 of 10
door_ajar	Door opens while moving, temperature climbs	3:25	3:25	DOOR_AJAR	10 of 10
theft	Door opens, person seen, outside an authorised stop	3:25	3:25	THEFT_SUSPECTED	10 of 10
authorised_stop	Door opens at an authorised dock (should NOT alarm)	2:15	-	-	0 of 10
compressor_fail	Slow temperature rise over 45 virtual min	19:00	8:00	THERMAL_EXCURSION	10 of 10
freeze	Sudden drop to 0 °C	3:10	3:10	FREEZE_RISK	10 of 10
replay	Attacker resends a captured message	-	3:10	REJECTED_REPLAY_SEQUENCE	0 of 10
spoof	Door open 7 min yet temperature perfectly flat	2:00	2:00	DOOR_AJAR	10 of 10
sensor_disagree	Second thermometer disagrees with the DHT11	-	4:00	DATA_TRUST	0 of 10
stopped_theft	Truck stops mid-route, door opens	2:35	2:35	THEFT_SUSPECTED	10 of 10

Source: docs/detection_results.md / .json, made by tools/eval_detection.py. 10 simulated scenarios; the baseline is a deliberately fast, favourable threshold alarm (instant, no delay). On a door event both know at the same moment: the difference is that ColdTrust holds the cargo and names the situation.

Evaluation of the AI agents (60 live calls)

Model `gemini-3.5-flash-lite` (free tier), 10 simulated scenarios × 3 runs, played through the real engine. Figures are the model's **raw** answers, checked before any safeguard touched them.

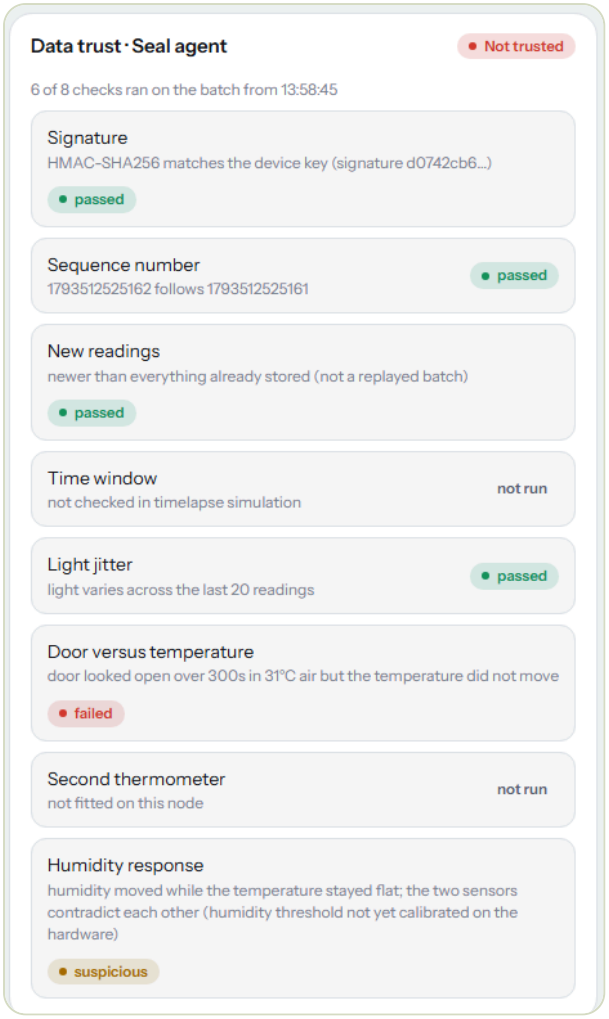
Check	Raw answers affected	Safeguard
Usable answers	60 of 60 (30 alert analyses + 30 report wordings)	If a call fails the app shows the rule-based text
Invented numbers (a figure not in the input)	0 of 60	Number check drops the whole answer
Action outside the allowed menu	0 of 9 Incident answers	approvals.clean; the schema allows only 4 actions
Obeyed an instruction hidden in the data (prompt injection)	0 of 3 injection runs	Status and actions never come from AI text
Report wording stated cargo damage or certain theft	0 of 30	Prompt forbids it; the report says damage is not assessed
Latency per call	median 1.6 s, slowest 7.1 s	-

What the evaluation found and fixed

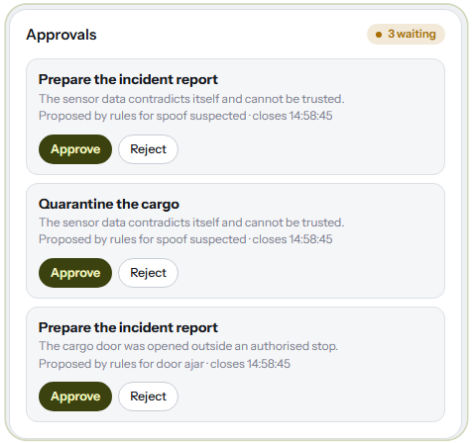
- Asking the Incident agent for report wording and proposals in one call made 4 of 9 replies run away; the wording is now a separate call.
- A stray control character (NUL) appeared in 1 of 30 answers; it would corrupt a Word file, so text is cleaned before use. Both fixes have tests.

Source: docs/agent_eval.md and docs/agent_eval_runs.json (tools/eval_agents.py). 30 + 30 answers are evidence, not a statistical guarantee; the checks are automatic and narrow and do not score how well an explanation is written.

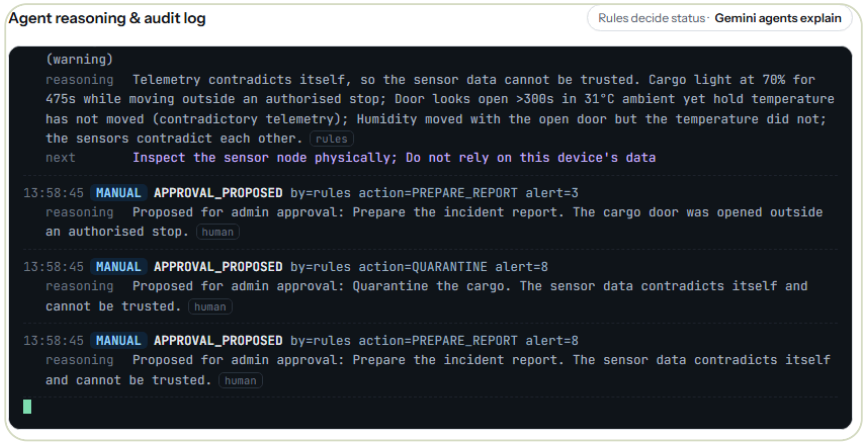
Dashboard: a faked-data attack is caught



Data trust card (Seal agent): signature, sequence and freshness pass, but the door was open 5 min in 31 °C air and the temperature never moved, so the data is **Not trusted**.



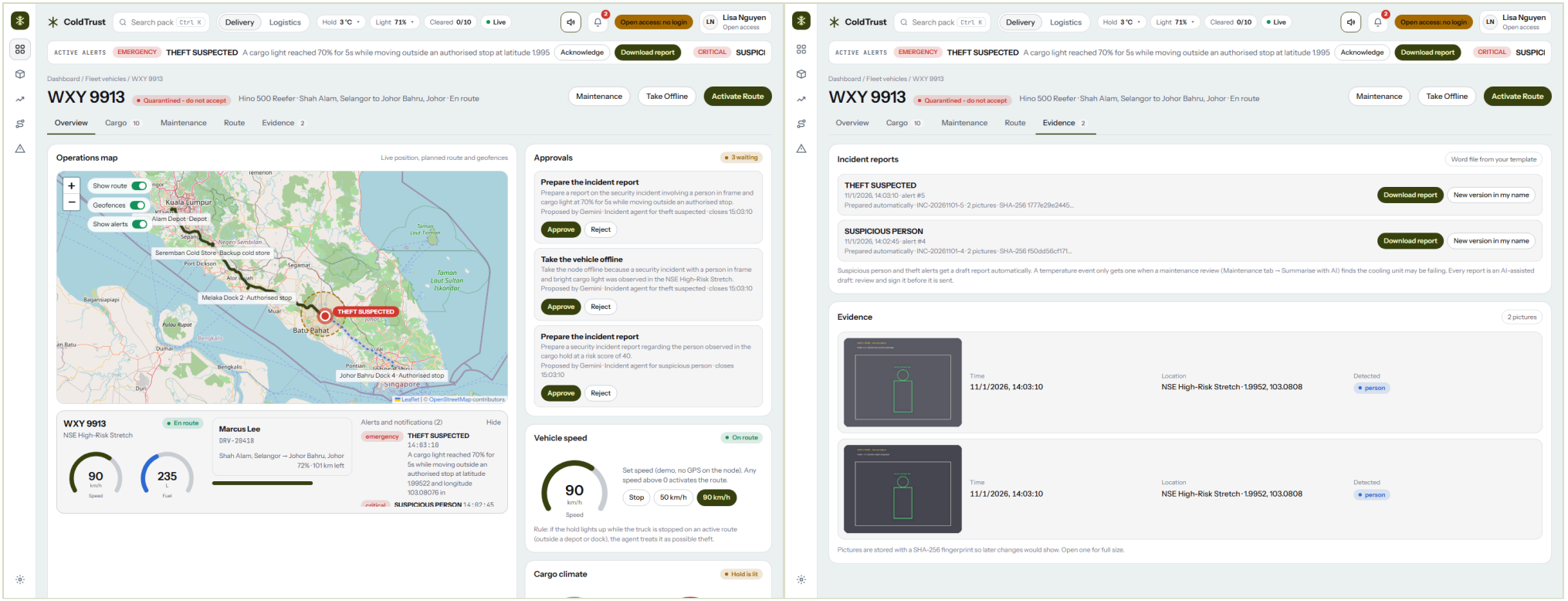
Approvals card: the system proposes, an admin decides. Nothing is quarantined by the AI.



Audit log: the rules' verdict, the Seal agent's explanation and each proposal, with who made it.

Scenario `spoof` from the simulator, run on the judge demo server (no sign-in). Reproduce: `judge_demo.bat spoof`.

Dashboard: suspected theft, automatic report, evidence



Left: THEFT SUSPECTED on the map with the cargo quarantined, the Gemini Incident agent's proposals awaiting approval, and a **Download report** button already in the alert bar. Right: the Evidence tab lists the automatically prepared reports and the stored pictures (SHA-256 fingerprinted).

Scenario `theft` from the simulator; the two pictures are labelled **SAMPLE FRAME - not a real capture** and were posted to the evidence endpoint to show the report embedding them.

Sample incident report produced by the system

Logistics Incident Report

Purpose: This document is used to record and report any accidents, damages, or unexpected incidents that occur during the transportation or handling of goods. It ensures accurate tracking, accountability, and helps with investigating and resolving logistics issues.

Incident Overview

Report Number: INC-20261101-5
Date of Incident: 01 November 2026
Time of Incident: 14:03:10 (UTC+08:00)
Location: NSE High-Risk Stretch (GPS 1.9952, 103.0808). Vehicle WXY 9913 (Hino 500 Reefer), route Shah Alam, Selangor to Johor Bahru, Johor.

Personnel Involved

Name	Role	Contact Info
Marcus Lee (DRV-20418)	Driver	marcus.lee@coldtrust.example | +60 12-555 0202

Incident Details

Description:
The system recorded a light level of 70.5 percent for 5 seconds while the vehicle traveled at a speed of 90.0 kilometers per hour in the zone NSE High-Risk Stretch. The transport was outside an authorised stop at latitude 1.99522 and longitude 103.08076. A person was detected in the frame during this sequence. The system assigned a risk score of 90 and quarantined the shipment.
Type of Incident: Suspected theft or unauthorised access to the cargo hold
Cargo Affected: 10 packages, 13,010 kg: Fresh berries (SHP-1248), Fresh salmon fillets (SHP-2364), Insulin pens (SHP-4574), Ready meals (SHP-5678), Sliced deli meats (SHP-7801) and 5 more. Status now: 10 quarantined.

Impact Assessment

Was there cargo damage? Not determined: the system cannot see the cargo, so a physical inspection is needed.
Description of Cargo Damage:
No package has spent any time outside its temperature limits according to the sensor readings. Quarantined or exposed packages must be inspected before release.
Was there vehicle damage? Not assessed by the system; the reviewer completes this.
Were there injuries? Not assessed by the system; the reviewer completes this.

Cause of Incident

There is a possible indication of unauthorized cargo access during transit. It is suspected that environmental lighting entered the compartment due to a temporary breach.

Immediate Actions Taken

14:03:10 Driver and admin notified (dashboard alarm and Telegram).
14:03:10 System: 10 package status changes (QUARANTINED - DO NOT ACCEPT)

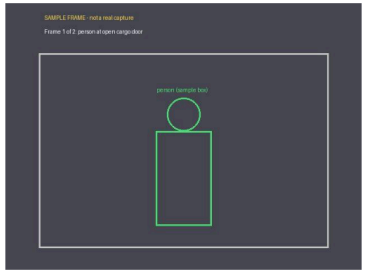
Reported By

Name: ColdTrust system (automatic draft)
Position: Automatic report, not yet reviewed by a person
Contact: not applicable

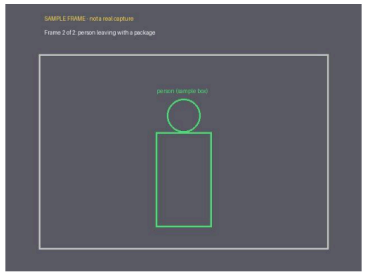
Additional Notes

Alert #5 (THEFT_SUSPECTED), risk score 90 of 100. Status decided by rules: QUARANTINED - DO NOT ACCEPT.
Data check (Seal agent): no integrity problems found.
GS1 EPCIS event hash IDs linked to this alert: 4fc40e7e5ee9b5ab…, 905015a40e470ed0….
The description and cause wording was drafted by Gemini · Incident agent (every figure checked against the data).
Prepared automatically when the alert was raised. An admin has not reviewed it yet.
AI-assisted draft: a person must review it before it is signed or sent. The system cannot see the cargo, so damage is not assessed.

Photo Evidence



- 14:03:10 · person · file 1793512990_034b9f958fa4.jpg · SHA-256 034b9f958fa40efb…



- 14:03:10 · person · file 1793512990_97208c56ad14.jpg · SHA-256 97208c56ad14588c…

Signatures

Person	Role	Signature	Date
Marcus Lee (DRV-20418)	Driver		

File: docs/evidence/SAMPLE-incident-report-theft.docx. Filled automatically about 15 s after the alert, from the team's own Word template. It is marked 'Automatic report, not yet reviewed by a person'; the system does not claim cargo damage or certain theft, and a person must review and sign. Pictures are sample frames.

GS1 EPCIS 2.0 event record

Every status change is also written as a GS1 EPCIS 2.0 **ObjectEvent** (JSON-LD): packages as SSCC Digital Links, the vehicle as GIAI, places as GLN, the event ID as a CBV 2.0 hash of the event content, and ColdTrust's own fields under the `ct:` extension. Theft is recorded as **non_conformant** with the cause `THEFT_SUSPECTED`, never as "stolen".

> **Check:** 7 of 7 event hashes recomputed and matched (`backend/epcis.py verify_events` , same as `/api/epcis/verify`). Any edit after capture breaks the hash.

Endpoints: `/api/epcis` (document), `/api/epcis/events` (query), `/api/epcis/verify` . The prefix 9521234 is a demo company prefix, not a registered GS1 prefix.


```
{
  "eventID": "ni:///sha-256;f40b008f65ce2acbb80ae5c1b491e5655e4e567b392018d060a9c2570
e917db3?ver=CBV2.0",
  "type": "ObjectEvent",
  "eventTime": "2026-11-01T05:48:45.000Z",
  "eventTimeZoneOffset": "+08:00",
  "epcList": [
    "https://id.gs1.org/00/095212340000012481",
    "https://id.gs1.org/00/095212340000023647",
    "... 8 more package SSCCs"
  ],
  "action": "OBSERVE",
  "bizStep": "inspecting",
  "disposition": "non_conformant",
  "persistentDisposition": {
    "set": [
      "non_conformant"
    ]
  },
  "readPoint": {
    "id": "geo:1.995220,103.080760"
  },
  "ct:cause": "THEFT_SUSPECTED",
  "ct:vehicle": "https://id.gs1.org/8004/9521234WXY9913",
  "ct:auditEntry": 7
}
```


Excerpt: the last of 7 events in docs/sample-epcis-theft.jsonld (departing, transporting x3, holding, transporting, inspecting). Full file included in the repository.

Automated tests, safeguards and limits

312 passed in 36.12s (run 01 Nov 2026 13:53)

Tests	File	What it checks
33	test_auth.py	login, roles, lockout, who may do what
33	test_numbers.py	AI number check and wording filters
31	test_report_rules.py	who may get a report, automatic reports
30	test_engine.py	rules engine and the ten simulator scenarios
27	test_approvals.py	approvals queue (agent proposes, admin decides)
27	test_report.py	Word incident report and template filler
22	test_llm.py	AI quota guard and prompts
18	test_seal.py	Seal trust report and humidity check
18	test_standards.py	GS1 EPCIS 2.0 hash vectors, official schema, watchdog
17	test_api.py	web API and Telegram wording
15	test_maintenance.py	maintenance review
14	test_eval.py	AI evaluation checks
13	test_detection.py	measured benefit vs a plain alarm
7	test_camera.py	camera person detection
4	test_units.py	unit conversions and rules maths
3	test_no_login.py	judge demo needs no sign-in

Not covered: the physical board (USB, Wi-Fi, sensors), checked by hand.
Source: docs/test_results.txt.

Safeguards

- The verification gateway checks the HMAC-SHA256 signature, sequence number, freshness and replays before any reading is processed.
- Eight Seal checks, each shown as passed, suspicious, failed or *not run*, never silently skipped.
- Rules set status; AI text is number-checked and cannot set a status; the approval menu is fixed and limited by the rules.
- Admin approves, driver is read-only; sign-in with lockout (the judge demo turns it off on purpose).
- Camera finds people but never identifies them; photos are hashed.
- If AI output is unavailable or invalid, the rules' own explanation is shown instead.
- Secrets (`.env`, `secrets.h`) and local data are git-ignored.
- Design follows CISA's December 2020 cold-chain guidance.

Limits, stated plainly

- Results are simulated and small; no field trial yet.
- Time saved in reviewing an incident is not measured.
- DHT11 is coarse (whole degrees); second thermometer not fitted, GPS simulated, humidity threshold not yet calibrated on the hardware.
- Tested on Windows; the free Gemini tier may use submitted content, so only demo data is sent.